1 A student is investigating the motion of a small cube on a turntable connected to an electric motor as shown in Fig. 1.1.

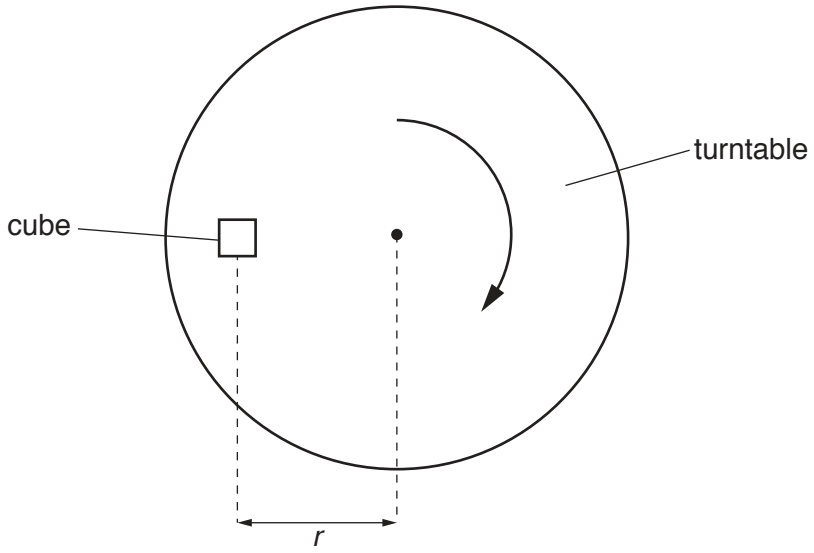


Fig. 1.1

The cube is placed at a distance r from the centre of the turntable. It is suggested that the relationship between r and the maximum frequency f of the turntable for which the cube does not move relative to the turntable is

$$K = 4\pi^2 mfr$$

where m is the mass of the cube and K is a constant.

Design a laboratory experiment to test the relationship between f and r. Explain how your results could be used to determine a value for K. You should draw a diagram, on page 3, showing the arrangement of your equipment. In your account you should pay particular attention to

- the procedure to be followed,
- the measurements to be taken,
- the control of variables,
- the analysis of the data,
- any safety precautions to be taken.

[15]

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